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Technical Report No. 39 (Revised 2021)

Guidance for Temperature-Controlled Medicinal Products — Maintaining the Quality of Temperature-Sensitive Medicinal Products through the Transportation Environment

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Guidance for Temperature-Controlled Medicinal Products — Maintaining the Quality of Temperature-Sensitive Medicinal Products through the Transportation Environment Team

Authors and Contributors

Chris J. Anderson, Cardinal Health

Rafik H. Bishara, Ph.D., RHB Technical Advising, LLC (Co-Chair)

Volker Cloos, Eli Lilly

Giovanni Curotto, veryQ pharma AG

Fabian De Paoli, GSK

Jean-Pierre Emond, Ph.D., The Illuminate Group

Michael English, Merck

Paul Harber, Parenteral Supply Chain, LLC

Nina Heinz, DHL

Zvonimir Majic, Ph.D., Teva

Mark Maurice, Sensitech (Co-Chair)

Janusz Mielewczyk, Amgen

Jahanvi Miller, MBA, PDA

Anthony (TJ) Rizzo, Cold Chain Technologies

Robert Seevers, Ph.D., Pearl Pathways

Dan Sorgen, Genentech

Joe Swigunski, Kite Pharma a Gilead Company

Cinthia Travaglini, UCB

Erik J. van Asselt, Ph.D., Merck, Sharp, & Dohme

Brian Wallin, Kite Pharma a Gilead Company

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1.0 Introduction

The distribution of life science products is extremely diverse, ranging from raw materials to finished product through the supply chain to the end user. Typically, as the distribution process moves from shipments on pallets to individual packages to delivery to the end patient, the term "Last Mile" is applied. The Last Mile defines the point at which products have reached their final destination (e.g., pharmacy, medical practice, private residence). Product temperature must continue to be maintained within registered storage and/or transport conditions. This may include shipments from a wholesale distributor to a pharmacy where the products are held awaiting customer pick-up or, as the Last Mile process has become an even more complex system, home delivery of products that also include cell and gene therapy.

1.1 Purpose

This technical report provides guidance for establishing a quality management system for distribution of all types of temperature-sensitive medicinal and pharmaceutical products, from preparation of products for transportation at shipping sites to the shipment of products to receiving sites for storage or distribution to patients (end users).

The examples accompanying the points to consider provided herein on how to protect the quality of the product are for demonstrational purposes only.

1.2 Scope

This technical report offers a resource for all stakeholders involved in the pharmaceutical supply chain and provides a model for qualifications from the process design through implementation and operation to verification. The stakeholders include, but are not limited to, the following organization types:

- Pharmaceutical and biotech companies of commercial and clinical medicinal products, including manufacturers, contract manufacture organizations (CMOs), and repackagers
- Health care providers (HCPs)
- Logistic service providers
- Wholesalers
- Distributors (including subcontractors)
- Pharmacies, including mail order, infusion, specialty, and compounding pharmacies
- 503B pharmacies currently in the United States
- Nongovernmental organizations such as the United Nations International Children's Emergency Fund (UNICEF) and Pan American Health Organization (PAHO)

The stakeholders referenced are responsible and accountable for the processes that control the handling, transport, and storage of the product from the point of manufacture to distribution to the patient. The stakeholders need to work closely with the manufacturers who have all the stability data and knowledge necessary to evaluate the excursion.

The content of this document follows the sequence of the model provided in **Figure 1.0-1,** covering process design, process qualification, and ongoing process verification for thermal protection systems in the pharmaceutical supply chain.

Note: Numerous technical reports from PDA and other organizations exist that provide more detail regarding temperature control management. These are listed in the reference **Section 9.0**.

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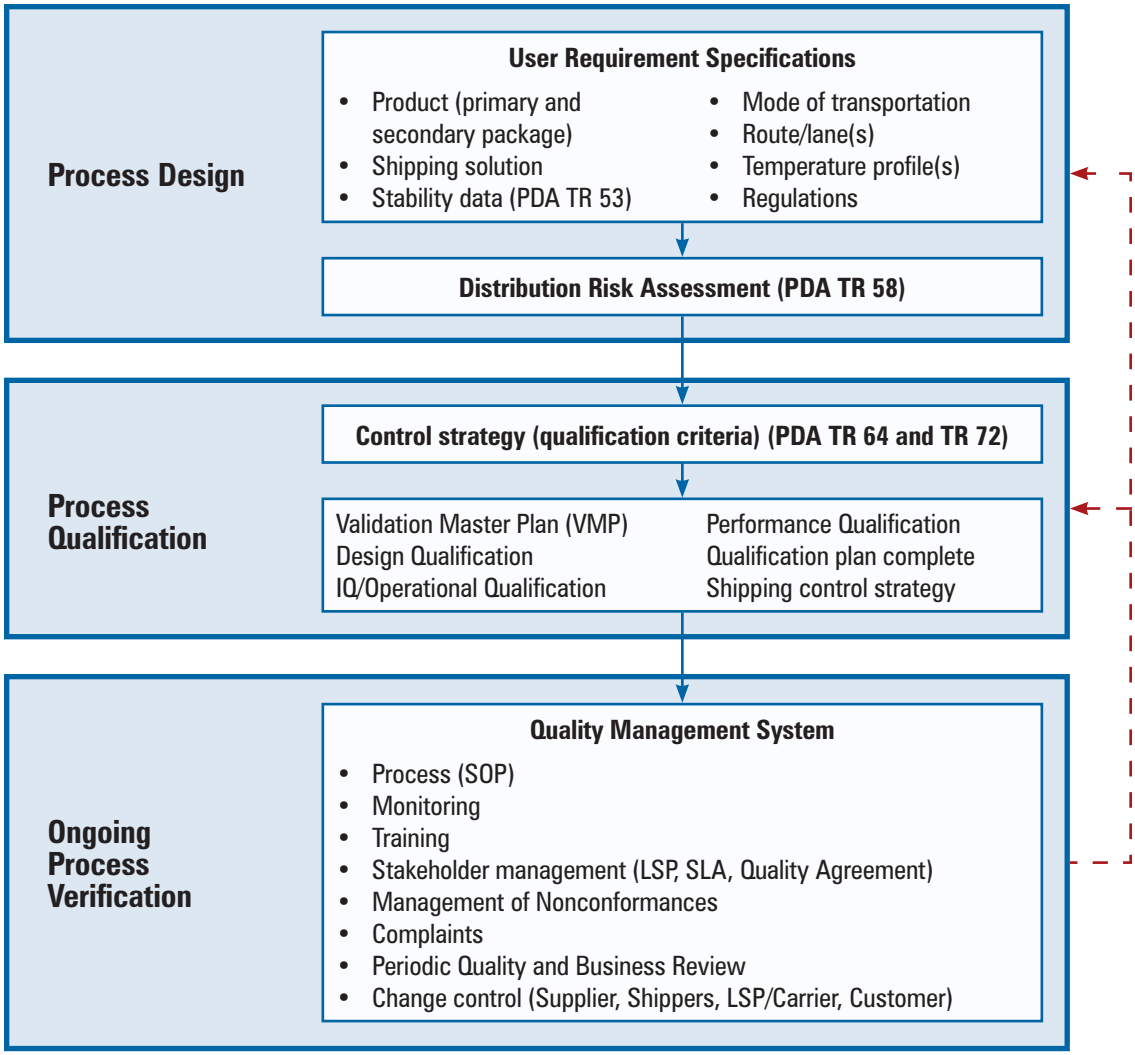

Figure 1.0-1 Qualification Process Flow and Key Deliverables

2.0 Glossary

The terms and definitions below are in common use within the industry and represent industry best practices in the experiences of the technical report team.

503B Pharmacy

The U.S. Food and Drug Administration (FDA) created this designation to, "recognize the differences between outsourcing facilities and conventional drug manufacturers, while maintaining the minimal standards necessary to protect patients from the risks of contaminated or otherwise substandard compounded drug products." US Pharmacies with this designation must comply with State Board of Pharmacy guidelines.

Acceptance Criteria

Predefined criteria that must be met by testing; numerical limits, ranges, or other suitable measures for acceptance of test results.

Ambient Temperature Profile

The temperatures selected to simulate the anticipated and experienced extremes used to challenge a thermal protection system in a testing chamber.

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Corrective Action/Preventive Action (CAPA)
A corrective action is taken to eliminate the root cause of a nonconformance and prevent recurrence. A preventive action is meant to eliminate the cause of a potential nonconformance or another potential undesirable situation (hence, aimed at preventing an occurrence).

Conditioning/Preconditioning
The process to bring the payload or a thermal protection system component(s) to a specified temperature condition by keeping it for a given period of time in the conditioning environment.

Data Acquisition System
A combination of sensors, measurement hardware, and software used to measure physical conditions that convert the results into digital numeric values. Data acquisition systems are used in thermal protection system qualification to measure temperature using, for example, thermocouples.

Data Loggers
Measurement instrument that is capable of recording temperature over a defined period of time based on memory capacity. The data can be retrieved, reviewed, and stored for accept/reject decision-making along with analysis of route.

Design Qualification (DQ)
Tests or assessments that are performed prior to operational qualification to ensure that the user requirements are met by the proposed thermal protection system.

Equilibration Time
The duration required for a thermal protection system with product load to reach the required product temperature range within the thermal requirements.

Indicators
A measurement instrument capable of comparing environmental conditions (e.g., temperature) to alarm settings to determine if those settings have been exceeded. Limited data may be stored.

Installation Qualification (IQ)
The process of obtaining and documenting evidence that the premises, equipment, and/or supporting systems have been provided and installed in compliance with their design specifications.

Logistics Service Provider (LSP)
Providers that contribute to the transportation and storage/distribution process including but not limited to carriers, integrators, forwarders, and custom brokers. LSP's may contract aspects of the logistics process to Subcontractors.

Management Review
A formal, documented review of the quality system's performance executed periodically at senior management level according to a planned schedule.

Mechanical Testing (also Distribution Testing, Physical Testing, or Dynamic Testing)
Testing that demonstrates and documents, with a high degree of assurance, the ability of a thermal protection system to withstand the distribution environment in compliance with the user and design requirements. Some typical examples are shock, drop, impact, compression, and vibration testing.

Nonconformance
Any unexpected and/or unwanted deviation from established processes, procedures, regulation, laws, requirements, standards, or specifications including but not limited to temperature, anomalous trends, and quality system failures.

Operational Qualification (OQ)
Testing that demonstrates and documents, with a high degree of assurance, that a system consistently performs as intended under the anticipated shipping conditions in compliance with the user and design requirements.

Pack-Out
The detailed instructions on how to precondition and assemble the packaging system with the payload to be shipped. With a seasonal pack-out configuration, the preconditioning and assembly may change depending on the time of year, ambient temperatures, and/or transportation lane. A universal pack-out configuration remains unchanged throughout the year.

Passive Thermal Protection System (see Thermal Protection Systems, Passive Payload (also Product Load))
The variable amount of product to be transported in the thermal protection system, which

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includes the combination of cases, cartons, and product assembled to be packed inside the vehicle. It may also refer to the volume of the space defined for product within the passive thermal protection system.

Performance Qualification (PQ)

Transport tests of the thermal packaging system and process that are conducted in the representative transportation or distribution environment by those who will be executing the day-to-day shipping process. The PQ serves to demonstrate that the qualified system is effective and reproducible.

Phase Change Material (PCM)

A physical material that absorbs energy when melting and releases thermal energy when freezing. PCMs are used in thermal protection systems to absorb or release heat energy from the external environment and maintain the desired temperature range of a packaging system. Dry ice is a type of phase change material. PCMs are also commonly called refrigerants, coolants, gel packs, bricks, and ice.

Placebo Product (also Mock, Simulated, or Dummy Product)

A placebo product is used to replace the actual product during qualification. Rather than use expensive pharmaceutical products, a placebo product provides a cost-effective way to test cold-chain solutions during the qualification process. The placebo product should imitate as closely as possible the dimensions and thermal behavior of the actual product and its packaging (primary and secondary). Preliminary tests may be necessary to prove that the placebo product is an acceptable emulation of the real product and its packaging (primary and secondary) that will be shipped in the qualified solution.

Proportional Integral Derivative (PID) Controller

The control loop mechanism that provides feedback on the performance of the thermal testing-chamber control system.

Qualification

Documented testing that demonstrates, with a high degree of assurance, that a process or system will function to meet its predetermined acceptance criteria.

Shipper

An individual or company that tenders products for transportation (common term also used to identify a package containing products to be shipped).

Stability Budget

A stability budget considers the results of long-term, accelerated, extreme excursion, and temperature cycling studies to determine the amount of time out of storage that a drug product may experience without any significant risk to its quality.

Standard Operating Procedure (SOP)

A written set of step-by-step instructions that describe the requirements necessary to complete an activity, task, or process.

Subcontractor

An entity that takes a portion of a contract from the principal contractor such as an LSP or from another subcontractor.

Temperature Excursion

An event in which product is exposed to temperatures outside of the recommended storage and/or transport temperature range.

Thermal Protection System (TPS)

Any active or passive temperature control system for the distribution of temperature-sensitive products.

Active Thermal Protection System

A temperature-controlled container with refrigeration and/or heating capabilities and its associated calibrated temperature-control devices. This term includes commonly used containers like temperature-controlled trucks/trailer, temperature-controlled ocean containers, and active-unit load devices for airfreight.

Passive Thermal Protection System

An insulated container that, when assembled, will provide thermal protection for its payload from outside temperatures. It comprises a set of components including an insulated container, spacers, dunnage, shipping refrigerants (e.g., gel pack, gel brick, dry ice), packaging, and product payload configured in a specific manner to function as a qualified temperature-controlled system for a specified duration.

Transportation Lane

The two points that define the origin and destination for a shipment.

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Transportation Route
The origin, in-transit, and destination points for a shipment.

User Requirements Specifications (URS)
Defines the users' critical parameters of the product, packaging, and transport system for the qualification process.

Validated Equipment
Validated equipment is used to conduct operational qualifications (OQ) including, but not limited to, thermal test chambers and data acquisition systems. In addition to OQ, equipment validation typically includes installation qualification (IQ), performance qualification (PQ), and periodic revalidation. Validated thermal test chambers should include an active preventive maintenance schedule. Data acquisition systems should include periodic calibrations to NIST or other accepted industry standards. For both validated and calibrated test equipment, the time between validation and/or calibration should be defined by predetermined time intervals documented by an internal procedure.

Validation Master Plan
A master plan is used to outline and define the validation scope, practices, and requirements that will assure the drug substance and drug product will retain all critical quality attributes as they move through their respective supply chains. This plan includes, for example, the qualification of the packaging, the choice between active and passive shipping containers, the route chosen, the weather and traffic conditions anticipated, and the selection and location of temperature monitoring devices.

2.1 Abbreviations

CAPA
Corrective Action/Preventive Action

CMO
Contract Manufacture Organizations

DQ
Design Qualification

HCP
Health Care Providers

IQ
Installation Qualification

LSP
Logistics Service Provider

NIST
National Institute of Standards and Technology

OQ
Operational Qualification

PAHO
Pan American Health Organization

PCM
Phase Change Material

PID
Proportional Integral Derivative

PQ
Performance Qualification

SLA
Service Level Agreement

SOP
Standard Operating Procedure

TPS
Thermal Protection System

TTI
Time Temperature Indicator

UNICEF
United Nations International Children's Emergency Fund

URS
User Requirements Specifications

VMP
Validation Master Plan

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3.0 Process Design

Process design begins with defining the requirements for shipping a product from the origin to the destination, which will require multiple inputs from development, logistics, etc. The outcomes of all requirements are a proposed thermal protection system (TPS) and the inputs for the transportation process qualification.

3.1 User Requirements Specification (URS)

The best practice for an organization is to complete a cross-functionally reviewed and approved User Requirements Specification (URS) before starting a qualification process (1). The URS defines the requirements of the product, packaging, and transport system for the qualification process. Examples of considerations may include but are not limited to:

- Product storage and transportation temperature
- Transportation method and/or lanes
- Performance duration required to perform activities such as pick, pack, ship, or put away
- Expected ambient temperature exposures or ambient temperature profiles
- Minimum and maximum expected shipping quantities
- Product load thermal characteristics and shapes (i.e., payload capacity)
- Temperature monitoring strategy
- Operational, safety, and logistical capabilities, including handling and material conditioning
- Environmental considerations (i.e., reusable, disposable, compostable, or recyclable)
- Other product specific considerations that should have been evaluated at the primary- or secondary-packaging level, such as shock, vibration, atmospheric pressure, are outside the scope of this document.

3.2 Quality Risk Assessment

Risk assessment is the process of identifying, evaluating, and mitigating risks that have the potential to negatively affect product quality, processes, and/or user/patient safety. It requires proactively identifying risks that have the potential to occur, evaluating their relative risk impact (using qualitative or quantitative scoring), and implementing effective design, process, and/or operator/user controls to minimize or eliminate those risks to product quality and user/patient safety.

Risk management procedures should be established to provide a standardized process to conduct, review, and update a risk assessment. A risk assessment is paramount to ensure that patient safety and product quality have been comprehensively considered and met through internal qualification testing, or equivalent vendor qualification testing.

Risk review sessions should be ongoing as well as event-driven (e.g., additional requirements, event-based failures, nonconformances, corrective/preventive actions (CAPA), changes). This will ensure previously evaluated risk levels are still representative of, or updated to reflect the current state of a product and/or process by means of the quality management system, which subsequently demonstrates the product and/or process is kept in an acceptable state of control.

The outcome of the risk assessment should be documented. Plans to handle, store, and distribute the product should be made accordingly. For more information about risk management, see PDA Technical Reports No. 54 and No. 58 and ICH *Quality Guideline Q9: Quality Risk Management* ***(2-4)***.

3.3 Design Qualification (DQ)

Design qualification (DQ) is the process of obtaining and documenting evidence that the TPS has been designed in accordance with user requirements. Principles discussed in this section apply to both active and passive TPS solutions.

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DQ testing may not be necessary for every process qualification, especially if the shipping system has already been developed and qualified for similar applications. If the risk analysis confirms the need for a Design Qualification, a protocol should be written. Results of the DQ should be documented.

When evaluating a supplier's prequalified TPS, the supplier's DQ does not need to be part of the evaluation package but may be leveraged to evaluate any process parameters, as long as they are technically justifiable and representative of the intended use environment.

DQ may be performed to ensure that the URS are met by the proposed solution; it is performed prior to OQ and performance qualification (PQ). The outcome of a successful DQ assures a high confidence level for successful OQ of a specific TPS. Typical process parameters may include, but are not limited to, the items listed in **Table 3.3-1**.

Table 3.3-1 Typical Process Parameters for Design Qualification

Description	Passive	Active
Process duration	Yes	Yes
Determining temperature monitoring device	Yes	Yes
Family approach rationale	Yes	Yes
Product quantity	Yes	Yes
Temperature-conditioning of product	Yes	Yes
Type of insulated material used	Yes	Yes
Product shipping configuration	Yes	Yes
Minimum and maximum thermal mass/product loads	Yes	Yes
Defined ambient temperature profile(s)—both hot and cold profiles, where appropriate	Yes	Yes
Temperature-conditioning of phase-change materials or active container	Yes	Yes
Location of refrigerants	Yes	No
Product used to stabilize temperatures such as phase-change materials	Yes	No
Orientation of package	Yes	Yes
Representative shipper selection strategy	Yes	Yes
Refrigeration system capacity	No	Yes
Air flow (forced)	No	Yes
Diesel or electric power supply	No	Yes
Operational modes (cycle sentry, continuous, front evaporator only)	No	Yes
Defrost cycles capability and inclusion	No	Yes
Power failure and recovery test	No	Yes
Loading and unloading pattern/time	Yes	Yes
Door opening	No	Yes

Hybrid active–passive systems also exist where all the items in **Table 3.3-1** may be applicable.

Design testing must consider both physical and thermal protection. The product transport evaluation should demonstrate that product can be transported without adverse impact to product quality attributes and package protection. Thermal qualification should demonstrate that a TPS can maintain the required temperature range and duration described in the URS.

Ambient temperatures within the shipping lane influence the capability of any TPS to maintain allowable payload temperatures. Service levels will affect the duration of exposure within a shipping lane.

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Initial selection and comparison of a potential TPS may often be made using a customized profile or one of several standard thermal profiles, e.g., International Safe Transit Association (ISTA) Testing Standard 7D or 7E ***(5-6)***.

Design testing results should demonstrate the best and worst-case locations to be monitored during OQ testing. Best practice considers using more than the minimum number of probe locations as described in **Section 4.0 Process Qualification**.

When a placebo product is used for testing, understanding its thermal dynamic compared to real products is crucial. For example, a box of 30 prefilled syringes of 1 ml will not behave exactly the same as a placebo product using a 30 ml vial of liquid. Assumptions used to design the placebo products should be well explained with preliminary test results or thermal calculations. The size of the placebo product should also correlate with the dimensions of the intended payload. In some cases, qualification will be done using minimum and maximum payload dimensions. In other cases, however, the use of a placebo payload with average dimensions is acceptable if most shipments have fairly constant payload sizes throughout the intended season(s).

DQ testing results are also used to define the components to be used as part of the TPS and in the OQ. Qualification results are only valid for the specific components tested. For example, an ice pack of a specific volume must also be described with the correct dimensions for a thermal packaging solution. An ice pack containing the same thermal mass but with a different shape or dimensions cannot be considered equivalent since a change of shape may have a substantial impact on the qualification results. Consequently, component specifications and diagrams should be written for each component of the successful design that is critical to the performance of the TPS. Component specifications are not required prior to design testing.

A qualification strategy for reusable temperature-controlled shippers should additionally include assessment on re-certified for use, recovery, and refurbishment services provided by the shipper, packaging vendor, or logistics service providers.

4.0 Process Qualification

The qualification exercise can be regarded as the deliberate and purposeful collection and evaluation of data from a process in order to establish evidence that the process can consistently satisfy the user requirements. The subtle difference between OQ and PQ is that the former is intended to examine the process at the extremes, and the latter is intended to ensure process technical transfer to the process owner. The OQ of a thermal protection system is typically executed in a climate chamber at the anticipated extreme conditions, whereas the PQ is executed in real life according to the Validation Master Plan. Collectively, these qualifications may be referred to as "qualification." A risk assessment should examine events that are exogenous to the process yet may impact product quality, such as equipment failures or unplanned shipping delays.

A qualified shipping system should have an approved set of qualification documentation. Typical parts of the documentation may include the URS, risk assessment, DQ, OQ, PQ, mechanical testing (for example shock and vibration), complete qualification plan, and final control strategy; (see **Figure 1.0-1**).

All qualification processes require a protocol and a report with a clear and precise description of how the tests were done, and the rationale for why specific procedures were selected (e.g., real or placebo

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products, air or liquid temperature). The methods section should describe precisely how the tests were prepared and completed, keeping in mind that anyone following those procedures should obtain the same results. Consequently, the methods section structure must:

- Describe the materials used in the tests
- Explain the conditioning procedure
- Describe the testing protocol
- Explain how measurements and calculations were achieved
- Describe, if necessary, which statistical analyses were used to analyze the data

Assembly diagrams and packing instructions are important to support the qualification process documents. The description of the preparations, measurements, and protocol should be organized chronologically. Photos documenting the assembly of the packaging systems (all components and products), location of temperature probes (if used), and experimental set-ups in environmental chambers should be included. Then, the protocol to be followed and a completed report should be approved by all parties involved.

The Validation Master Plan (VMP) encompasses all aspects of validation required by an organization. The validation may include, for example, manufacturing processes or equipment, back-up power equipment, heating and cooling systems, storage, and distribution. The goal is to make sure that all items related to temperature management for storage and distribution have been identified, included in the VMP, analyzed and documented, and have produced acceptable results.

Thermal qualification should not use a macroscopic approach (whole or lumped system without internal spatial variations), which can be misleading. Rather, the thermal qualification should be obtained by performing a heat transfer analysis targeting the detection of product temperature excursions at critical locations, such as corners and points nearest or farthest from a cold pack in a passive TPS or refrigeration unit, for an active system. Critical product temperature locations can be determined using preliminary test results, thermal simulation software, or other guidance from documents such as *ISTA Standard 20 (Revision 2.0): Design and Qualification of Insulated Shipping Containers (ISC)* ***(9)***. It means that, in most cases, more than one temperature measuring point is needed for the qualification process.

4.1 Operational Qualification (OQ)

The goal of an OQ is to provide documented testing that demonstrates, with a high degree of assurance, that a process or system will function to meet its predetermined acceptance criteria when exposed to anticipated environmental extreme conditions (for example temperature, shock, and vibration). The package or system would be run according to the preapproved qualification protocol with expected loads to prove reliability and repeatability of performance. The location of products or placebo products within the shipper must be clearly documented. Including only the number of products or items is not sufficient; describing their orientation and positions with respect to one another (loading configuration) is essential since it will influence the thermal behavior of the product payload and the temperature distribution within it.

The equipment used for an OQ should mimic the expected temperature conditions within the lane and record accurate data of the environment and product payload. The materials to be used in the test must be preconditioned in the controlled environments defined in the OQ protocol. The protocol may include but is not limited to a description of:

- Scope and activities required to complete an OQ test
- Clear acceptance criteria
- Preconditioning of the test materials
- Temperature-monitoring methodology

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- Ambient temperature profile to stress the active or passive solution
- Product payloads
- Shipper orientation
- Physical protective properties, which is often tested separately
- Rationale of all tested conditions explained and documented

Mechanical performance testing should also be performed as part of the OQ unless otherwise justified. The test plan should be representative of the actual shipment conditions, such as use of the subcontracted resources, and ensure product quality is maintained during transit. The results should be documented in a report. It may prove beneficial for the end user to also create a separate summary of the OQ testing results for internal approval.

In case the OQ process requires the use of an environmental chamber to perform the selected temperature profiles, it is essential to consider, that not all environmental chambers will provide the same testing conditions, even though they can replicate the same temperature profile. Such parameters as product placement and stacking pattern, airspeed, shelf surface types (e.g., grid, metal, wood, plastic), and temperature accuracy will have a significant impact on test results and should be given considerable attention. The same also applies for the qualification process of potential handling scenarios such as package orientation, temporary storage conditions, and stacking before shipping. For example, most small parcel shipments will experience multiple orientations during handling and should be tested to at least identify potential temperature excursions that were not observed during the original design orientation. Such observations should be communicated to the design team for possible design revaluation.

4.2 Performance Qualification (PQ)

Performance qualification (PQ) is done using the actual transportation or distribution environment and is performed by those who will be executing the day-to-day shipping process. The PQ serves to demonstrate that the qualified system is effective and reproducible in real-world application. Testing should be performed using typical parameters including product loads, operational handling, transportation routes, and conditioning procedures.

PQ testing should be performed following a preapproved protocol and the results should be presented in a report. The test plan should incorporate standard operating procedures (SOPs) that include:

- Actual ambient temperature variances, including seasonal changes and cross-hemisphere lanes, if applicable; data should be available to support the lane
- Representative transportation load configurations
- Defined packaging configurations
- Calibrated temperature monitors
- Sufficient number of tests to assure reliable results
- List of SOPs relevant to the process
- Evidence of operator training on the relevant SOPs (extending to all subcontractors)

Other procedures may be used and, if so, they need to be justified.

For further understanding, see **Section 8.0 Case Studies.**

4.3 Qualification Plan Complete

To close the qualification of the TPS, all completed documentation and the deviations should be summarized in one report, sometimes called Qualification Assurance Summary Report or Validation Summary Report. The conclusion should indicate the applicability and shortcomings of the TPS and list the deviations from the URS. The report should be approved according to established procedures.

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4.4 Shipping Control Strategy

A shipping control strategy or quality risk control strategy, which is the output of the quality risk assessment, for drug product in the supply chain should define the measures to implement the qualified TPS within the operational environment to ensure that the product is shipped under the intended conditions within the stability budget. This means anything that can increase the degradation rate of the medicinal product must be controlled. A control strategy should include the implementation or update of the VMP, SOPs, contracts (e.g., quality agreement, technical agreement), training, and maintenance protocols including calibrations. In addition, a control strategy may include, in part, allowable times and temperature exposures expressed in a tabular form that reflects product stability, citing the stability budget for the product. This will ensure the qualified TPS will be implemented and used according to the established qualified conditions.

5.0 Ongoing Process Verification

After process design and process qualification of a TPS, the shipping organization should implement the TPS in operations according to the control strategy. By applying continuous process verification, this will ensure that the TPS will protect the product throughout transit—from its preparation at the shipping site, during transport, and until its arrival at the receiving site—as intended during the qualification for quality, effectiveness, and safety.

Continuous process verification applies to all transportation modes and to all parties involved in the transportation of medicinal products and its related activities, such as (but not limited to):

- Preparing the goods for transport
- Loading/unloading goods into shipping equipment
- Loading/unloading from one piece of shipping equipment to another
- In-transit handling between transportation (e.g., airport transit, harbor transit)
- Receipt of goods

Each participant in the distribution process has an obligation to support the goal of protecting the product, minimizing opportunities for exposure to temperatures outside the required range, and to assess and respond appropriately based on the impact of such exposures, should they occur. In particularly cold or hot climates, it may be advisable to extend the temperature protection to the docking areas of the vehicles (complete enclosure)

Participants in the transportation process can be categorized as shippers, logistics service providers (LSP) and receivers. LSP refers to the providers that contribute to the transportation and storage–distribution process, including carriers, integrators, forwarders, and customs brokers.

There are also considerations specific to transportation using vehicles or containers providing active temperature-controlled environment as well as specific aspects for transportation using passive TPS.

In order to support implementation, shippers must thoroughly document the processes used and improvement modifications made, as well as the reasoning for those changes, during the qualification process. This documentation is the critical reference for the creation of a process SOP, which should include procedures for process monitoring, environmental monitoring, nonconformance management, training, and the application of these procedures by the managing stakeholders.

Process monitoring must demonstrate that the state of control is being maintained where the set of controls consistently provides assurance of continued process performance and product quality. The

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effectiveness of lane assessment should be evaluated to establish the state of compliance to deliver products with the expected quality required by customers, patients, healthcare professionals, and regulatory authorities.

Monitoring tools, control systems, and regular trend analysis must be in place to measure performance and establish the level of distribution process capability.

Distribution process capability should be checked based on risk management and organizational SOPs to facilitate the identification and evaluation of opportunities for correction and improvement.

Transfer of the qualified process to the operational areas and the involvement of service providers (stakeholders) is formally accomplished by means of approved procedures. When those procedures are followed, the controls of critical process parameters will ensure in repeatable, successful transport of medicinal products. The written, approved procedures should be in place, and responsible personnel should be trained prior to implementation of the process. Consideration should be given to the procedures attached to the OQ and PQ reports.

Relevant personnel and organizations participating in the temperature-controlled process should be provided instruction concerning the principles of packaging, qualifying, and transporting temperature-sensitive medicinal products. The purpose of the training is to ensure that the shippers, transport service providers, and their subcontractors maintain the integrity of the transport chain for medicinal products.

5.1 Standard Operating Procedures (SOP)

SOPs should be in place to describe in detail all steps of the process execution. SOPs should be reviewed and revised periodically according to the organization's policies and guidelines. A key part of an SOP implementation would include training and a comprehension evaluation section to assess the knowledge of the SOP users (e.g., shippers, logistic service providers, carriers). A summary of changes and versioning is required for easier tracking. The review and approval system requires a defined process of retrieval and removal of obsolete SOP versions.

5.2 Monitoring

Data loggers and indicators, commonly referred to as monitors, are instruments that use microprocessors with sensors and internal memory capacity to collect data on predefined attributes, such as temperature and/or humidity. They can be connected to a computer system and use software to collect, store, and analyze data as, in an example of temperature-controlled storage premises, part of a continuous monitoring requirement. In such cases, computer/electronic system validation is required as a part of the GxP.

In different circumstances, monitors can be connected wirelessly to a modem, device, or cloud-based system, or monitors can operate independently as stand-alone instruments. Data loggers can record and transmit predefined attributes in a real-time environment providing certain predispositions are fulfilled, such as connectivity to a GSM/Wi-Fi/Bluetooth signal operated within the restrictions of airfreight transport. Unlike temperature, humidity monitoring and control is predominately reserved for storage premises. Like shock and vibration, humidity monitoring during transport is a product-specific consideration, not a standard in attribute monitoring.

When supported by the risk assessment, a time–temperature indicator (TTI) can be used as an alternative to data loggers. A TTI is a device or smart label that shows the accumulated time–temperature history of a product. TTIs are commonly used to indicate exposure to excessive temperature and/or

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excessive time above or below a defined temperature. TTIs are usually smaller than data loggers and can assist with immediate decision-making due to a clear symbol or color visible on the device.

The number and position of monitors used in a shipment should depend on the qualification studies and control strategy. Conducting a risk assessment is recommended to determine the use of monitors on each shipment, as distribution qualification cannot simulate all external environmental conditions that can influence the shipment's temperature profile. The monitor position in a TPS should be determined by conducting a worst-case monitor-position study according to the temperature-mapping qualification data. An external data logger may be used to support the root cause analysis of a nonconformance.

The frequency of monitoring or measurement interval should be based on a risk assessment that encompasses regulatory guidelines and requirements and includes, but is not limited to:

- Shipment duration
- Interval or frequency mandated by country regulations
- Memory capacity
- Stability budget of the product

If used, temperature monitoring during transportation should reflect the capability of the transportation provider and the TPS to maintain the product shipment within the acceptable temperature range as defined on the product label or allowed by the manufacturer. Monitors should be started, at the latest, as the container (e.g., lorry, truck, trailer, sea and air container, passive shipper) is closed and, in turn, should be stopped when the container is opened at the destination. Start-up delay can be used to account for the time required for loading and equilibration.

During picking, staging, loading, unloading, and put away of temperature-sensitive product, conditions should be controlled to preserve the quality of the product according to its stability budget.

Each monitor is required to have a certificate of calibration and must not be used beyond the date specified on the certificate. The expiration of the calibration should be based on the sensor manufacturer's recommendation and/or local regulations.

5.3 Training

Training on all SOPs should be completed by all those who will be responsible for implementing the task the SOPs describe. Training records should be maintained and updated promptly so they are complete and available for internal Quality Unit audits and external regulatory inspections.

5.4 Stakeholder Management

The management process for vendors or service providers is an important component of a quality management system (QMS). Vendors, for the purpose of this document, are defined as qualified contract acceptors and their approved subcontractors. The contract-acceptor approval process comprises verification of the vendor's QMS system and level of compliance to good distribution practices (GDP). Verification for direct contract acceptors (providers of the logistics or transport service) is typically performed through an audit or comparable assessment process based on risk assessment (as per organization's procedures). The frequency of quality and GDP audits for logistics vendors is related to the type of the service they are providing, based on risk as defined in the organization's procedures.

The type and scope of the contract would vary, depending on the type of service required, but the contract would require a signed quality agreement or quality service level agreement, in conjunction with the business agreement, as a part of the approval process. Not all LSPs require an audit performed

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directly by the organization using them. Thus, the most important elements of the vendor approval process are:

- Audit
- Quality Agreement or Quality Service Level Agreement
- Key performance indicators (KPI) or performance metrics for continuous improvement verification

A review of KPIs should be conducted periodically. Ad-hoc or for-cause audits could be required based on poor performance or significant changes related to services offered in accordance with an organization's internal procedures.

The main areas of a vendor's QMS verification depend on the scope and services offered. Typically, a quality or GDP audit, as applicable, might include the following:

- Organization, staff, training
- Quality management system
- Deviations/nonconformances
- Supplier assessment/auditing program
- Record management (paper and electronic)
- Change control
- Annual reports
- Facilities, vehicles, and equipment, including environmental monitoring and qualification
- Cleaning, sanitation, hygiene, pest control
- Maintenance, including calibration
- Deliveries
- Warehouse operations and qualification
- Safety

5.5 Management of Nonconformances

From a temperature perspective, distribution must be designed to keep the product within the defined limits described by the manufacturers, applicable regulations, or visible on the outer packaging. The distribution process is a complex and dynamic process, where product is exposed to the external environment, less controllable than e.g., a typical drug manufacturing process, not managed directly by the product manufacturer and exposed to various entities in the supply chain. Due to the inherent variability of the external environment, nonconformances should be expected and documented according to the level of impact on the product.

Written procedures should be in place so that nonconformances are defined, their impact on product quality is assessed, and their root cause investigated, determined and, as applicable, eliminated (see **Section 5.5.1**).

5.5.1 Categorization of Nonconformances

Nonconformances can be categorized by their type (e.g., temperature excursions outside the defined limits described by the manufacturer or visible on the outer packaging, logistic process deviation, equipment breakdown, theft/loss, anomalous trends, noncompliance with a good-practices requirement), as well as their criticality (e.g., no impact or potential or determined minor or major impact on product quality or patient safety). The categorization applied by each company must be described in a written procedure.

5.5.2 Impact Assessment

An impact assessment must be performed following written procedures. The impact assessment must also consider and define the boundaries of the nonconformance in terms of which products may have been affected and which are out of scope.

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The level of impact determines the level of investigation and correction needed. A prompt correction of the nonconformance to limit its immediate impact is required, and all actions undertaken in that respect must be properly documented.

All products impacted (or potentially impacted) must be individually assessed. An investigation should be conducted involving the appropriate subject matter experts (SME) for the product(s) to determine the impact on product quality. Stability data might be used to assess temperature excursions. Nevertheless, the cumulative impact of any possible previous excursion must be assessed. All investigations including temperature excursions should be properly documented and approved by the Quality Unit.

When product quality, compliance, or patient safety is not impacted, it is considered acceptable to close the nonconformance with minimal or no root cause investigation (as applicable and described by the procedure). The impact assessment must be formalized and approved as required by the local procedure.

5.5.3 Corrective and Preventive Actions (CAPA)

If a root cause of a nonconformance is determined, a CAPA should be considered and implemented in order to remove the root cause of the nonconformance. A corrective action is taken to prevent recurrence, whereas a preventive action is taken to prevent occurrence. Depending on the CAPA, the implementation of the CAPA should be monitored for its timeliness and effectiveness. The thoroughness of a CAPA must be commensurate with the risk posed by the nonconformance.

A CAPA is used to address quality and compliance issues and should result in process improvements and process understanding. It should be used in conjunction with the root cause analysis. To facilitate the trend analysis, a root cause should be identified according to the elements, for example, the route where the nonconformance occurred, root cause location (where it happened), and root cause owner (who or what is at the origin). The level of effort, formality, and documentation of the actions should be commensurate with the level of impact of the issue on product quality.

The development, implementation, and verification of the CAPA's effectiveness will often require collaboration with the LSP(s) and/or packaging/container manufacturers.

5.5.4 Product Control and Disposition Decision

Should an absence of impact for products in the boundaries of the nonconformance be determined, the product can be distributed after that no-impact determination has been confirmed, according to product specifications and local procedures.

Should an impact be determined, product(s) must be held and can be distributed only after a correction of the impact, if possible, has been successfully executed and product quality impact no longer exists.

When correction of the impact is not possible, impacted product(s) must not be released.

5.5.5 Approval and Closure

The responsibilities for approving nonconformances and their investigations must be described in a local procedure. The Quality Unit is typically responsible for approving all nonconformances as described.

The person responsible for the approval and closure of a nonconformance must have access to all information and investigation elements as required to perform the closure. When more information or investigations are needed to enable the approval and closure of a nonconformance, these must be requested from and provided by the investigation team.

The approval can be granted only after all investigative elements are available and formalized and all applicable CAPAs have been satisfactorily identified and approved.

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5.6 Complaints

Similar to nonconformances discovered during distribution operations, procedures should be documented that describe the processes for receiving, evaluating, investigating, addressing, and trending of complaints from customers. A CAPA should be considered to remove root causes of complaints as appropriate.

Products returned due to complaints must be segregated and prevented from subsequent distribution, A comprehensive product quality impact assessment must always be performed by the manufacturer, if product is planned to be further distributed, as defined by local procedure and allowed by local regulation.

5.7 Periodic Review

Distribution performance must be monitored and reviewed periodically to ensure the state of control is maintained as expected by the qualifications. The extent and frequency of the distribution process capability should be reviewed at regular time intervals using a risk-based approach.

Distribution nonconformances (e.g., number, trends) and the system for the management of nonconformances must be periodically reviewed. The aim of the review is identifying negative patterns or trends in the processes and highlighting areas for improvements.

Implementing a process to monitor the occurrence of nonconformances, by tracking appropriate metrics and KPIs, is necessary. Mechanisms for escalation should be included to discuss the trends.

Aspects to be reviewed include, but not limit to:

- Number or frequency of nonconformances for each process or product, as applicable
- Adequacy (effectiveness) of CAPAs (e.g., by measuring recurrent nonconformances)
- Adherence to the standards required by the local procedure (e.g., timeliness for investigating and approving nonconformances)

For nonconformances classified as minor, trending should be performed to ensure they do not represent (or hide) a major nonconformance that would warrant more incisive actions.

This periodic review must be an integral part of the management review.

5.8 Change Control

An effective change control system must be in place to control changes made to the qualified systems. Each change to a qualified system must be described and documented.

According to the type of change, different systems can be used to record and manage changes (e.g., distribution change control, system logbook).

In distribution, a good practice is to maintain a list with all the routes/shipping lanes and their description (e.g., shipping/receiving site, LSP, airlines/haulers, cold chain solution).

Changes made to the system may render the distribution process no longer qualified. All changes (planned or unplanned) and their potential impact on the qualification status must be documented and approved by the Quality Unit.

Examples of criteria that could trigger a change control process might include:

- New transportation lane with a specific transportation mode (e.g., air, ocean, rail, road) or major modification of an existing qualified transportation lane (e.g., shift from air to ocean)
- New type of data logger
- New passive TPS

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Minor changes can be followed by a periodic report shared during management reviews.

- Minor modification of an existing qualified transportation lane (for example change of the transit airport)
- Change of the alarm/alert limits of an existing data logger

In contingency situations, such as an element for pandemic, proper risk assessment and change control should be assessed.

6.0 Conclusion

This technical report provides guidance for establishing a process for the qualification of distribution of temperature-sensitive medicinal and pharmaceutical products. It offers a resource for all involved stakeholders in the pharmaceutical supply chain, providing a model for qualifications from the process design through implementation and operation to verification including the shipment preparation of the products and thermal protection systems, their transportation and arrival at the receiving site for storage and further distribution to the patient (end user).

The technical report addresses "what" is needed to qualify thermal protection systems and processes, but not the "how" as this may vary depending on local regulations, business requirements, and product characteristics. By applying the guidance provided in this technical report, the reader will ensure that product quality is maintained during the distribution through the pharmaceutical supply chain in compliance with current regulatory expectations.

7.0 References

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8. *Van Asselt, E. J., Zvonimir, M, Bishara, R. H., and Stener, E.: Qualification of Pharmaceutical shipping Systems in seven Steps. Pharmaceutical Manufacturing and Packaging Source (PMPS), May 2020 Edition, pp.56 – 59.*
9. International Safe Transit Association. *ISTA Standard 20 (Revision 2.0): Design and Qualification of Insulated Shipping Containers (ISC)*; ISTA: East Lansing, Mich., 2014.

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8.0 Case Studies

These studies have been included as examples of an approach to Qualification of storage and transport systems, but not to the exclusion of other approaches.

8.1 Case Study 1: Distribution Incident Trends Analysis

CAPAs should be created for major and critical distribution deviations that have an impact on the product quality, for example, temperature excursions with product impact or product damages.

For events and minor deviations that happen in the distribution environment, CAPAs are not mandatory, as there is no impact on product quality and integrity as defined by the documented procedure.

The objective of monthly trend analysis is to confirm CAPAs opened for major and critical deviations and to prioritize CAPAs according to destinations with the highest reported number of minor deviations.

For every deviation reported, follow the checklist on page 19 (**Figure 8.1-1**).

8.2 Case Study 2: Passive Shipper SOPs

When using passive shippers, the assembly diagram, photos, and a description should be included within either the SOP or protocol. As with a puzzle, the multiple pieces must be assembled correctly. A written description of the process to assemble a passive shipper must accompany the diagram. Training should be completed and documented for all employees who will assemble passive shippers. Retraining schedules should be developed and maintained. Audits need to be performed on the accuracy of the passive shipper assembly. Random checks may be done after a passive shipper has been assembled and finalized. The Quality Unit should randomly select shippers and open them to verify proper assembly.

Transport modes need to be defined prior to the execution of any qualification. A carrier quality agreement may be helpful to understand exactly how logistic service providers and freight forwarders will consistently handle each shipment. Things to consider are the route and the risks with that route related to time in transit and/or security.

Gel pack and phase-change materials are used inside passive shippers to maintain temperature. Frozen gel packs and phase-change materials can be at different temperatures, such as -15 °C to -10 °C. Knowing at what temperature these materials should be, according to the passive shipper requirements, is essential to understand. Consistency is needed to assure that equipment used to condition and store gel packs and phase-change materials are capable of producing gel packs and phase-change materials at the proper temperature. The next step is to determine how long gel packs and phase-change materials need to stay inside the equipment to ensure the proper temperature. In most cases, it may require many days to condition these products depending if they are stored on a pallet, in a box, or individually.

The goal is to make sure the materials are at the correct temperature prior to use. Using gel packs and phase-change materials prior to being at the proper temperature could result in failed shipments. If gel packs and phase-change materials are used frozen, it is imperative to determine whether or not they will cause irreversible damages to the payload. Gel packs and phase-change materials may freeze if they are come in direct contact with products affected by freezing (thermal shock and denaturing the product); this should have been tested during prior qualification processes. Thermal shock can occur unintentionally if the cold bank moves during handling and ends up touching the payload, or if the shipper is placed in a cold room prior to shipping. The layout and written plan should include a process to warn users that the cold bank may require material to separate it from the payload. Introducing gel packs and phase-change materials at the wrong temperature into a pack-out or improperly placing them can result in adverse temperatures.

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- ❑ Determine if the reported deviation is relative to damage, temperature, or another reason (e.g., logistic split)
- ❑ Determine the root cause based on gathered information
- ❑ Document the root cause according to the following information:
  - ❑ Type of freight (air/sea/road/rail)
  - ❑ Site of origin
  - ❑ Site of destination
  - ❑ Logistic service provider
  - ❑ Root cause owner (e.g., may be handling agent, hauler, airline, cold chain solution supplier)
  - ❑ Root cause location (e.g., may be storage or tarmac time at transit airport, motorway rest area, aircraft)
- ❑ Monthly review
  - ❑ Perform a monthly review report based on a rolling 12-month period of distribution deviations
  - ❑ Calculate per site destination the percentage of deviations (= number of deviation/ total shipments)
  - ❑ Sort the deviation percentage (by decreasing order)
- ❑ Identify the site destinations with the highest deviation percentage.
  - ❑ Based on a fixed percentile (e.g., Percentile 95, meaning the 5 % of destinations with the highest deviation percentage), count the occurrence for same root cause location and owner
  - ❑ If more than one occurrence, identify the occurrence as a cluster of a potential issue to investigate further
  - ❑ If occurrence is more than two within the 12-month rolling period (destination/ root cause location/owner), investigate further and create a CAPA accordingly
- ❑ Biannual review
  - ❑ Perform a biannual review based on the last 12 months of distribution deviations
  - ❑ Calculate per root cause owner and per root cause location the number of deviations/shipments
  - ❑ Sort the deviation percentage by decreasing order
  - ❑ Identify the root cause owner and root cause location with the highest incident percentage
  - ❑ For the top three root cause owners and root cause locations, investigate further and create a CAPA accordingly

Figure 8.1-1 Deviation Checklist

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8.3 Case Study 3: Qualification Master Plan

The following case study is an example of how to implement the information contained in this technical report. Risk analysis and planning for its mitigation should be the first step.

To make validation terminology applicable to a good distribution practices (GDP) environment, a qualification master plan (QMP) is written to define the qualification strategy. For each qualification step (design, operational, and performance qualifications), a rationale is written to support the qualification program according to the type (active or passive) of thermal protection system (TPS).

More specifically, when the qualification program addresses winter and summer conditions in the operational qualification (OQ) and performance qualification (PQ), particular attention is given to the PQ done with commercial batches that can last one year or more.

Performing a TPS PQ with consecutive shipments of the product is considered a best practice, because it mimics the real situation and therefore confirms the representative OQ result. However, considering the recommendations from the U.S. Food and Drug Administration and the World Health Organization, which advocate the PQ be completed before commencing commercial distribution of the product, it is necessary to document and develop a rationale to ensure the product quality used during the PQ.

After the OQ, an intermediate qualification summary report is written, based on the OQ conclusion, to justify the decision to use commercial product for the PQ. The output of the OQ performed considering winter and summer conditions is to evaluate temperature risk during static conditions (thermal chamber temperature profiles) and performance of the solution in order to define the number of probes and locations that have to be used for routine operations. In order to demonstrate that the OQ results are effective and reproducible over the years, PQ field transportation tests are performed in real shipping conditions (dynamic condition), during which the temperature is monitored and documented in a PQ winter and PQ summer report. PQ will consist of replicated live transportation tests done as defined in the intermediate validation report. Based on a matrix approach, multiple destinations and product loads may be considered. The number of replicate(s) should be governed by the regulatory requirements of the specific country or region. Studies should be done in winter and summer conditions at the time of the year where the weather and route are challenging to assure the results are representative. Regional anomalies should be taken into consideration e.g., during extreme weather conditions; monsoon.

Another option to consider is having a continuous performance verification program in place, where all shipments are monitored, and the performance of a temperature-controlled shipper is verified based on temperature data and the number of nonconformances experienced on regular quarterly/annual basis. Deviation wording may need to be changed, as root cause analyses are mainly performed for major and trending nonconformances. All minor nonconformances, by nature, do not have product quality or patient safety impact and therefore do not require a root cause analysis or implementation of corrective and preventive actions (CAPA).

The output of the PQ is to evaluate temperature during real shipment and performance of logistics processes with the TPS.

Until the final Quality Assurance Summary Report is approved, in addition to the PQ shipments, real transportation shipments will be made based on the output from the OQ validation results. There is no product risk, as every shipment is temperature-monitored in a location defined by the analysis of the thermal mapping results obtained during the OQ phase. In addition, any temperature nonconformance will be documented. Based on the risk analysis and mitigation policy of the company, a root cause analysis will be conducted, for example, for major excursions or trends, to guarantee that the transportation system is under control.

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A final Quality Assurance Summary Report is issued once the PQ is completed and the qualification process is closed.

9.0 Additional Reading

ANVISA – Brazilian Health Regulatory Agency
- ANVISA Guide 02: Transportation system

ASTM – American Society for Testing and Materials
- ASTM F2825 Standard Practice for Climatic Stressing of Packaging Systems for Single Parcel Delivery: Packaging System

BPOG – BioPhorum Operations Group
- BPOG Best Practice on Transport Qualification: Thermal Protection System

EC – European Commission
- European Commission GDP: Containers AND "qualified equipment (e.g., thermal packaging, temperature-controlled containers or temperature-controlled vehicles)"

FDA – Food and Drug Administration
- U.S. Food and Drug Administration. *21 CFR Part 11, Electronic Records; Electronic Signatures—Scope and Application*; Pharmaceutical cGMPs. Government Publishing Office: Washington, D.C., 2003.

HDA – Healthcare Distribution Alliance

HC – Health Canada
- Health Canada (GUI-0069): Shipping Container

IATA – International Air Transport Association
- IATA Perishable Cargo Resources – Chapter 17 Air Transport Logistics for Temperature – Sensitive Healthcare Products: Active container solutions OR Active Systems AND passive packaging systems

International Council for Harmonisation
- *Quality Guideline Q10: Pharmaceutical Quality System*; ICH: Geneva, 2008.

ISO – International Organization for Standardization
- ISO TC122: qualified equipment like thermal packaging, temperature-controlled containers or temperature-controlled vehicles

ISPE – International Society of Pharmaceutical Engineers
- ISPE Good Practice Guide – Cold Chain Management: Portable Packaging or Shipping Systems
- International Society for Pharmaceutical Engineering. *GAMP Guide: Records & Data Integrity*; ISPE: Bethesda, Md., 2017.

ISTA – International Safe Transit Association
- ISTA PCG-01 Revision 00 – Reusable Passive Thermal Packaging Best Practice Guideline

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- ISTA PCG-02 Revision 00 – Passive Thermal Packaging System Operational Qualification Best Practice Guideline

PDA – Parenteral Drug Association
- PDA Technical Report No. 46: Last Mile: Guidance for Good Distribution Practices for Pharmaceutical Products to the End-User
- PDA Technical Report No. 52: Guidance for Good Distribution Practices (GDP\s) for the Pharmaceutical Supply Chain: Qualified Packaging Container
- PDA Technical Report No. 53: Stability Testing to Support Distribution of New Drug Products
- PDA Technical Report No. 64: Active Temperature-Controller Systems: Qualification Guidance
- PDA Technical Report No. 72: Passive Thermal Protection Systems for Global Distribution – Qualification and Operational Guidance: Passive Thermal Protection System

USP – United States Pharmacopeia
- <UPS> 659 – Packaging and Storage Requirements
- U.S. Pharmacopeial Convention. General Chapter (1079) Risks and Mitigation Strategies for the Storage and Transportation of Finished Drug Products. In *USP43–NF38* USP: Rockville, Md., 2020; p 7635.
- <USP> 1160 – Pharmaceutical Calculations

WHO – World Health Organization
- WHO DIP 818 Guidelines on the international packaging and shipping of vaccines: Shipment Unit
- WHO Annex 9 Model Guidance for the Storage and Transport of Time- and Temperature-sensitive pharmaceutical products: Shipping Containers AND insulated passive containers AND Active Containers

White Papers
- Van Asselt, E. J., Zvonimir, M, Bishara, R. H., and Stener, E.: Qualification of Pharmaceutical shipping Systems in seven Steps. Pharmaceutical Manufacturing and Packaging Source (PMPS), May 2020 Edition, pp.56 – 59.
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Guidance for Temperature-Controlled Medicinal Products — Maintaining the Quality of Temperature-Sensitive Medicinal Products through the Transportation Environment

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Bethesda Towers
4350 East West Highway
Suite 600
Bethesda, MD 20814 USA
Tel: +1 (301) 656-5900
Fax: +1 (301) 986-0296
E-mail: info@pda.org
Web site: www.pda.org